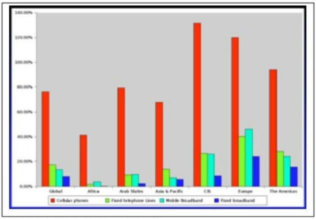

Figure 3: Cell phone and broadband users

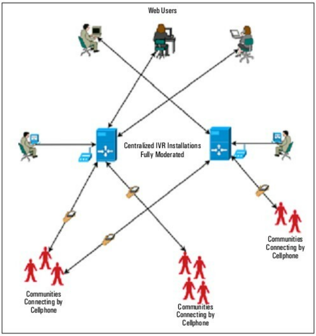

Figure 4: Centralised deployments

are now actively engaged in developing technology to allow people to use their voice to connect themselves and their communities to the rest of the world. One of the first tools of this new age of innovation is the audio portal.

An audio portal?

An audio portal (Figure 2) is essentially a website with a lot of audio content that can be accessed both through the Web as well as by phone. While the Web interface is usually like a blog, the phone interface is an IVR (Interactive Voice Response) system, where users press keys to navigate through menus and content. In more advanced IVR systems, voice recognition may be used, though this is still limited to the well-documented accents of the English language. The Web interface is very similar to a blog, and several audio portals do use the blog layout. Behind the scenes, the platform will also provide an interface to manage posts. Early implementations of audio portals tended to rely on specialised moderation consoles, which have media-previewing capabilities as well as functionality for moderators to add meta-data, such as a summary and title, to the content to make it friendlier to users on the Web.

Users will typically call the IVR interface to record and listen to content using their cell phones, while Web users will access the website interface to listen to the audio posts using a browser, and leave comments in text, which then may or may not be converted to audio using a text-to-speech system.

People who own the latest Android or iPhone may find the idea of an IVR interface to browse content somewhat counter-intuitive, since it makes no sense to call in and scroll through a set of menus, particularly with an irritatingly monotonic voice rattling out instructions all the time, when you can simply open the Web page on your cell phone's browser, and read. The graph in Figure 3 may help clarify why a purely visual interface is not adequate to reach the majority of the world. The percentage of Internet users, even among the mobile phone users of the world, is a fraction of the percentage of people using their phones purely for voice and SMS. While mobile Internet use is, and will continue to be, on the rise, the bulk of the world will continue to be on voice for some time to come.

This is also historically consistent, since most societies have far stronger oral traditions than written ones. Voice captures much more than simply language. Tone, quality, emotion are all interwoven in the spoken word. If a picture is worth a thousand written words, then a spoken word counts for at least a few hundred... not to mention that drawing an attractive picture takes considerably more skill than speaking!

What makes mobile phones particularly attractive as a medium, though, is the two-way nature of the medium. With radio and television, though the reach may be much wider than mobile phones, the ability to respond immediately to what you hear or see—on the same platform, at the same level as the source, which is extremely valuable in fostering dialogue—is missing. The audio portal concept caters to every cell phone, whether mass-market or smart-phone equally, which works very well to level the platform. Most importantly, audio portals use technology, skills and other resources that are available now, as opposed to those that require extensive 'capacity building' exercises. This is probably the reason why audio portals, as a tool, find more favour with grassroots workers and members of the community, rather than with technology evangelists and academia.

The technology

Audio portals utilise relatively simple technology, most of which has been around in the open source world for some time. An audio portal will typically consist of a phone interface (either fixed-line or mobile), connected to a content-management system (usually a database) and a Web front-end, via an IVR running on a soft switch or software PBX system. Two examples of audio portal platforms are Swara and FreedomFone.